

A Pipelined CISC Processor: the 68040

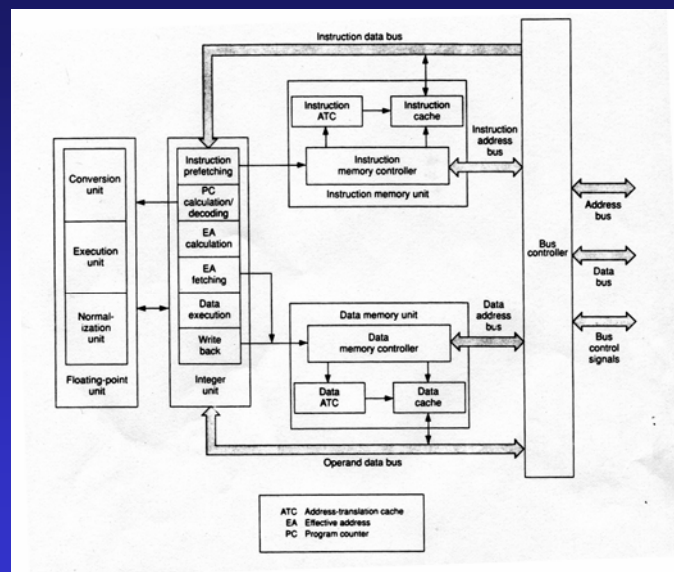
- The Motorola 68040 CPU dates from 1990
- It has
 - Separate Instruction and Data Caches on-chip
 - a 6 stage integer pipeline
 - a 3 stage floating point pipeline
- It is a CISC processor
 - primarily because of the need to maintain backwards compatibility with earlier 68xxx processors
- In its time it was a highly successful processor
 - it was used by Apple, NeXT, and other workstation designers
 - there are still a few in use
 - which is not bad for a 14-year old design!
- It was the last mass-produced processor in this series
 - (although there was a 68060: the series was replaced by the PowerPC range)

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The 68040 outline architecture



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The 6 68040 integer pipeline stages

- Instruction Prefetch
 - fetches instructions in blocks of 128 bits
- PC calculation and instruction decode
 - takes 48 bits, decodes instruction, and can process 16, 32, or 48 bit instructions (including immediate operands)
- EA calculation
 - calculates EA for operands. This stage is internally microcoded.
 - some EA calculations require more than one cycle.
- EA fetching
 - may perform a data memory access
- Data execution
 - performs operations, transferring results to write-back buffer
- Write results back
 - updates register set or data memory

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Notes on this architecture

- All the accesses to memory...
 - whether data memory or instruction memory
- ...are actually cache accesses
- This is critical to permit 1-clock cycle memory access
- Note the structural hazard between the EAF and WB stages for access to the data memory
 - This is the result of the CISC instruction set.

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Branches in the 68040 pipeline

- The CPU always computes the branch-taken address
- It speculatively starts both the branch and branch not taken instructions for conditional branches
- Special shadow registers are used allowing both instructions to start, but to be able to be undone safely.
- The pipeline is optimised for the branch-taken case
 - Motorola did dynamic tests on software, and reckoned this was the more frequent case.